

Port Scanning: SYN, Connect, UDP & Timing

Port states mean something

Open: service listening, accepting connections

Closed: port reachable, no listener (useful: reveals host is alive)

Filtered: firewall or ACL drops your packets

Scan types

```
# SYN stealth (needs root) – leaves no full connection
sudo nmap -sS -T2 10.0.0.20 -oN stealth.txt

# TCP Connect (no root needed) – full handshake, logged by service
nmap -sT -T3 10.0.0.20

# UDP (slow, noisy) – find SNMP, DNS, TFTP
sudo nmap -sU --top-ports 50 10.0.0.20

# Comprehensive (all ports, versions, scripts, OS)
sudo nmap -sS -sV -sC -O -p- --min-rate 1000 10.0.0.20 -oA full
```

Timing and IDS evasion

```
nmap -T1 target    # paranoid slow – avoids most IDS
nmap -T5 target    # insane fast – loud, likely blocked
nmap --scan-delay 100ms target  # pace to avoid rate limits
nmap -f target     # fragment packets (less modern IDS worry)
```

How services are detected

Banner grabbing: the service's "hello" string (ssh, smtp, ftp banners)

nmap probes: after finding open, sends service-specific probes

service and version (-sV) detection: fingerprint libraries vs. behavior

What a big scan reveals

Exotic ports (3389 RDP, 5900 VNC) = privileged access by mistake

Different OSes = possible dev/test servers forgotten

Multiple versions = upgrade gaps and decision debt

"Scan fast for hosts, scan narrow for ports, scan deep on interesting ones." A professional scan respects time budgets and target fragility.

[← Module start](#)

[Service Enumeration Deep →](#)